QuiverBioscience

Below are a list of drug discovery/development-related prompts/queries that an serious drug developer or executive would ask an AI-enabled drug discovery tool for the CNS, such as Sapphire:

- **<u>Examples from pitch deck:</u>**
    - Are there known disease-causing variants in Gene X that could guide a splice-modulating or allele-selective ASO strategy?
    - What is the predicted minimal knockdown threshold needed for phenotypic rescue/efficacy based on human neuronal phenotypes?
    - What ASO target sites on this transcript show the highest predicted efficacy? lowest toxicity?
    - Does the gene require selective targeting to avoid unwanted biological consequences?
    - Based on known oligo distribution maps, will an ASO targeting this gene achieve sufficient exposure in the relevant brain regions or neuron classes?
    - What other disease-causing genes produce similar electrophysiological or transcriptomic rescue signatures? Can they be addressed by the same oligos?
    - What is the patent landscape for ASO target sites on this gene (target coordinates, sequence claims, modification claims)?
    - Does this transcript display features that increase probability of success? (e.g., long half-life, accessible RNase H sites, favorable exon/intron architecture
    - Evaluate the risk of pro-inflammatory cytokine release (IL-6, TNF-alpha) for this specific chemical modification pattern.

- **<u>Original Examples from James/Graham:</u>**
    - Which gene perturbations are most similar to TSC2 based on their EP profiles/signatures?
    - Do the TSC1 and TSC2 gene perturbations have close proximity?
    - Which gene perturbations are anti-podal to TSC1/TSC2 based on their EP profiles/signatures?
    - Please list all of the genes that are “hits”
    - Which gene perturbations are most severe in their disruption of neuronal function?
    - Which voltage-gated sodium channels cluster together?
    - List the gene perturbations within the mTOR pathway and confirm enrichment or identify mTOR pathway genes whose signature is not aligned with those of the other mTOR members.
    - How internally consistent are electrophysiological signatures among gene perturbations in the mTOR pathway, and which outliers deviate most strongly from the cluster?
    - “Within the mTOR signaling pathway, which gene perturbations produce electrophysiological profiles that are least similar to other members of the pathway?”
    - Are genes associated with the EMC Complex present in the Quiver gene perturbation data set?
    - If so, do the EMC Complex genes cluster together? If yes, how many and which ones?
    - Rank KEGG pathways by the internal similarity of their gene perturbation profiles.”
    - Are EP profiles from the PI3K-Akt pathway distinguishable from those of the Wnt signaling pathway?
    - Can we reconstruct the KEGG mTOR pathway structure from EP similarity alone?
    - Do epilepsy-associated gene perturbations cluster more closely together than a randomly selected set of 50 gene perturbations?

- **<u>True Drug Discovery Prompts to dictate external dataset prioritization:</u>**

- **TARGET DISCOVERY & PRIORITIZATION**
    - Rank the top 25 genetically supported targets for disease modification in early Alzheimer’s disease based on Quiver functional phenotype reversal, human genetics strength, and biomarker measurability.
    - For TSC2 loss-of-function, identify all gene perturbations in Quiver that are functionally antipodal. Rank them by druggability and BBB feasibility.

  - Identify CRISPR hits that normalize excitatory/inhibitory imbalance in human cortical neurons and overlap with schizophrenia GWAS loci.
  - For SCN2A gain-of-function epilepsy, propose target suppression strategies ranked by predicted functional rescue and safety margin.
  - Map all genes functionally converging on the same Quiver embedding cluster as GRIN2B and identify which are more druggable.
  - Identify gene targets in neuroinflammation that show strong functional rescue in human neurons but minimal microglial toxicity signals.
  - Among 18k gene perturbations, which 50 genes most strongly improve neuronal firing stability without reducing network connectivity?
  - Identify pathway “common nodes” across ASD, schizophrenia, and epilepsy functional clusters.
  - For Parkinson’s disease, identify targets with protective genetic variants and strong Quiver phenotype modulation.
  - Rank top 10 CNS targets where Quiver functional signal contradicts transcriptomic predictions.

- **ASO DESIGN PROMPTS**
  - Design 5 optimized ASO sequences targeting TSC2 with predicted >70% knockdown efficiency and minimal off-target risk.
  - Generate allele-specific ASOs for a pathogenic gain-of-function mutation in SCN8A.
  - For MECP2 duplication syndrome, design partial suppression ASOs that achieve 30–50% knockdown without complete silencing.
  - Identify antisense targets predicted to normalize hyperexcitability in Dup15q neurons.
  - Rank ASO chemistry modifications (2’MOE, LNA, PMO) by predicted CNS tolerability for gene X.
  - For gene Y, generate ASO sequences optimized for intrathecal delivery with low innate immune activation risk.
  - Predict toxicity risk for the top 10 ASO candidates targeting gene Z using known ASO safety datasets.
  - Propose combination ASO strategies for polygenic epilepsy clusters.
  - Identify antisense approaches that replicate the functional phenotype of a known protective LoF human variant.
  - For rare monogenic neurodegeneration gene A, design ASOs and estimate IND timeline feasibility.

- **SMALL MOLECULE DISCOVERY**
  - Identify FDA-approved molecules that are functionally antipodal to gene X LoF phenotype.
  - Rank small molecules in Quiver with strongest rescue effect in TSC2 neurons and BBB penetrance evidence.
  - Propose novel scaffolds structurally similar to compound cluster Y but predicted to improve selectivity.
  - Identify molecules that normalize E/I imbalance without suppressing intrinsic excitability.
  - For Nav1.7/Nav1.8 dual inhibition, propose a small molecule profile predicted to avoid cardiovascular liability.
  - Generate 3 novel chemical structures predicted to reverse disease phenotype cluster Z.
  - Identify repurposing candidates with prior Phase 2 CNS safety data and strong Quiver phenotype match.
  - Compare small molecule vs ASO strategy for gene X in terms of predicted PoS and rNPV.
  - Identify compounds in the atlas with disease-modifying phenotype but low historical clinical investment.
  - Rank 20 CNS compounds by predicted Phase 2 probability of success using Quiver + external data

- **COMBINATION THERAPY**
  - Identify drug combinations that produce additive rescue in ASD-like functional cluster.
  - Propose combination strategies targeting excitatory neuron intrinsic excitability + inhibitory synaptic transmission.
  - Identify combination pairs that maximize phenotype rescue while minimizing toxicity signals.

- For treatment-resistant depression, propose dual-mechanism combinations supported by Quiver embedding similarity.
- Identify synergistic pairs for SCN2A epilepsy predicted to outperform monotherapy.

- **TRANSLATIONAL & BIOMARKER QUESTIONS**
  - For gene X, identify fluid biomarkers that align with the Quiver functional phenotype.
  - Which CRISPR perturbations correlate best with known EEG phenotypes in epilepsy?
  - Map Quiver excitability signatures to clinical symptom domains in schizophrenia.
  - Identify functional signatures predictive of cognitive improvement vs mood improvement.
  - For neurodegeneration gene Y, predict whether functional rescue implies disease modification or symptomatic effect.
- **PORTFOLIO & CAPITAL ALLOCATION**
  - Rank top 10 CNS programs by probability-adjusted NPV using Quiver + external competitive landscape.
  - Identify targets with high functional signal but low competitive saturation.
  - For gene X, estimate peak sales potential and regulatory risk profile.
  - Identify programs likely to fail in Phase 2 based on weak functional-human alignment.
  - Recommend which 3 programs should receive $50M allocation for maximum PoS improvement.

- **STRATEGIC DUE DILIGENCE**
  - For target X, how many companies are currently active and what stage are they in?
  - Identify adjacent pathway members that are more druggable than the current target.
  - Evaluate whether targeting gene X is likely to face safety liabilities based on historical toxicology.
  - Compare ASO vs small molecule approach for gene Y in terms of regulatory path and commercial scalability.
  - If we were building a first-in-class CNS franchise, which Quiver-derived target cluster provides the highest long-term strategic value?

- **MECHANISM DISAMBIGUATION**
  - For gene X, does the Quiver phenotype more closely resemble synaptic dysfunction or intrinsic excitability shift? Quantify effect sizes.
  - If we suppress gene Y by 50%, what compensatory pathways are predicted to activate?
  - Identify targets whose CRISPR phenotype is reversed by known anti-inflammatory drugs.
  - Which gene perturbations cluster with known rapid-onset antidepressants?
  - For target Z, is the functional phenotype likely upstream or downstream of mTOR signaling?
  - Identify genes whose perturbation worsens phenotype in rodent-supported neurons but improves in human-only systems.
  - Does the Quiver signature of gene X suggest disease modification or symptomatic modulation?
  - Identify targets that normalize spike waveform morphology but not firing rate — what does this imply mechanistically?
  - Rank perturbations that stabilize network synchrony without suppressing firing frequency.
  - Which CRISPR hits produce phenotype reversal only under synaptic assay, not intrinsic excitability assay?

- **GENETICS + FUNCTION INTEGRATION**
  - Identify genes with strong GWAS support but weak Quiver phenotype.
  - Identify genes with strong Quiver phenotype but no GWAS support — are these novel opportunities?
  - For protective LoF variant carriers in gene X, simulate expected electrophysiology profile.
  - Identify gene clusters converging on common functional embedding despite different genetic pathways.
  - Map ClinVar pathogenic variants onto Quiver perturbation space.

  - Identify genes where partial knockdown gives superior functional outcome compared to full knockdown.
  - Rank LoF vs GoF perturbations for gene Y by predicted clinical feasibility.
  - Identify polygenic convergence points in ASD functional clusters.
  - Which targets show strong effect in excitatory neurons but minimal effect in inhibitory neurons?
  - Identify genes whose perturbation mimics effect of top 3 antidepressant compounds.

- **BBB, PK/PD, AND DRUGGABILITY**
  - For top 20 Quiver targets in schizophrenia cluster, rank by BBB penetrance feasibility.
  - Identify targets predicted to require >80% CNS exposure to show effect.
  - Which targets are likely extracellular and antibody-accessible?
  - For gene X, simulate required brain exposure for functional rescue.
  - Rank top 30 hits by small-molecule ligandability.
  - Identify targets with structural homology enabling virtual screening acceleration.
  - For molecule cluster Y, predict metabolic liabilities.
  - Identify ion channels with strong functional signal but cardiovascular risk.
  - Compare oral small molecule vs intrathecal ASO for gene Z — which gives higher therapeutic index?
  - Identify targets with narrow therapeutic window based on Quiver amplitude-response curves.

- **TOXICITY & SAFETY PREDICTION**
  - Predict seizure risk liability for top 15 excitability-modulating compounds.
  - Identify gene suppressions predicted to cause hypoexcitability-related cognitive impairment.
  - Rank ASO sequences by predicted innate immune activation.
  - Identify compounds whose Quiver profile resembles known neurotoxic agents.
  - For target X, estimate likelihood of psychiatric adverse events.
  - Predict long-term synaptic destabilization risk.
  - Identify genes essential for neuronal survival in baseline state.
  - Which compounds reduce firing rate below physiological range?
  - Rank gene perturbations by risk of developmental toxicity.
  - Identify targets whose suppression induces compensatory upregulation of oncogenic pathways.

- **COMBINATION & NETWORK STRATEGY**
  - Identify synergistic gene-gene suppression pairs.
  - Predict dual-target small molecule profiles for ASD cluster.
  - Identify combinations that normalize E/I ratio without affecting total spike output.
  - Rank combinations by predicted reduction in Phase 2 failure.
  - Identify gene suppression + small molecule combination strategies.
  - For SCN2A epilepsy, simulate combination of partial knockdown + sodium channel modulation.
  - Identify pathway node where single intervention produces maximal downstream correction.
  - Predict multi-target strategy outperforming mTOR inhibition alone in TSC.
  - Identify hub genes whose perturbation collapses multiple disease clusters.
  - Rank top 5 two-target combinations for depression based on functional rescue + safety

- **RARE DISEASE & PRECISION**
  - For monogenic epilepsy gene X, design ASO achieving 40% knockdown.
  - Identify genotype-specific rescue strategies.
  - Predict response heterogeneity across iPSC donor lines.
  - Identify targets for pediatric intervention with minimal developmental risk.

- Map newborn screening targets to functional clusters.
- Predict severity gradient from electrophysiology amplitude.
- For ultra-rare gene Y, is there sufficient effect size to justify IND?
- Identify targets suitable for orphan designation.
- Rank rare disease targets by time-to-IND feasibility.
- Identify CRISPR perturbations rescuing severe LoF phenotypes.

- **COMMERCIAL & COMPETITIVE INTELLIGENCE**
  - For target X, list all companies in Phase 1–3.
  - Identify white-space opportunities in schizophrenia MoA landscape.
  - For gene Y, is market saturation high?
  - Estimate peak sales for top 5 ASO programs.
  - Rank targets by exclusivity durability.
  - Identify MoAs crowded by generics.
  - Predict pricing potential for rare genetic CNS target.
  - Identify targets with low competition but high genetic support.
  - Compare mTOR vs adjacent pathway target commercialization potential.
  - For portfolio of 10 programs, optimize for $10B revenue in 10 years.

- **PORTFOLIO OPTIMIZATION**
  - If budget is $100M, allocate across 5 programs for max rNPV.
  - Kill one of these three programs based on data.
  - Rank current CNS portfolio by translational robustness.
  - Identify 3 programs to fast-track to IND.
  - Identify 3 programs to terminate preclinical.
  - Predict Phase 2 PoS uplift from adding Quiver validation.
  - For $50M investment, which cluster maximizes PoS delta?
  - Compare small molecule platform vs ASO platform ROI.
  - Optimize portfolio for 30% PoS improvement.
  - Simulate outcome if top 2 programs fail — what is backup plan?

- **AI SELF-REFLECTION & UNCERTAINTY**
  - Where is Quiver data insufficient to support this hypothesis?
  - Identify assumptions driving ranking.
  - Provide uncertainty intervals around predictions.
  - Which predictions rely most heavily on external data?
  - Where does transcriptomics contradict electrophysiology?
  - Identify predictions most sensitive to model drift.
  - Which targets have weak replication across plates?
  - What additional experiment would most increase confidence?
  - If we had 2x power in screen, which targets likely emerge?
  - Provide confidence score for ASO sequence prediction.

- **EXTREME STRATEGIC PROMPTS**
  - If you were CSO at Lilly, which 3 targets would you license?
  - If you were CSO at Roche, which disease-modifying target is most compelling?
  - For Pfizer scale, which target yields $5B+ potential?
  - For BioMarin rare focus, which gene is best monogenic candidate?

- For Alnylam RNAi model, which targets are most suppressible?
- For Regeneron antibody-first strategy, which extracellular CNS targets stand out?
- If we aim to dominate CNS in 15 years, which cluster is franchise anchor?
- Identify a first-in-class opportunity with minimal competition.
- Identify a best-in-class improvement opportunity.
- What is the single highest-value program in the entire Quiver atlas?

- **Monogenic & Rare Neurogenetic Disorders**
  - Find a clinically de-risked drug for SCN2A gain-of-function epilepsy using Quiver EP rescue ranking.
  - Identify ASO strategies for MECP2 duplication syndrome and rank by predicted functional rescue and toxicity profile.
  - Find small molecules that reverse SYNGAP1 haploinsufficiency phenotypes using multi-modal rescue scoring.
  - Identify clinically de-risked drugs for Fragile X Syndrome based on EP + transcriptomic rescue alignment.
  - For CDKL5 deficiency disorder, rank FDA-approved CNS drugs by EP rescue score.
  - Identify combination therapies for STXBP1 encephalopathy that maximize E/I balance restoration.
  - For KIF1A-associated neurological disorder, identify suppressor targets from Quiver's CRISPR atlas and rank druggability.
  - Identify allele-specific ASOs for SCN8A gain-of-function variants.
  - For Rett syndrome, find mTOR-adjacent mechanisms that provide safer alternatives to rapalogs.
  - Rank anti-seizure drugs by predicted long-term network stability in TSC2 KO neurons.
  - For Dravet syndrome (SCN1A LOF), identify drugs antipodal to inhibitory neuron dysfunction.
  - Identify synaptic stabilizers for Phelan-McDermid syndrome using Quiver E/I metrics.
  - For Angelman syndrome, identify rescue strategies targeting UBE3A pathway nodes.
  - Identify BBB-penetrant small molecules correcting CACNA1A-associated phenotypes.
  - Rank antisense vs small molecule strategies for GRIN2B-related neurodevelopmental disorder.
  - For PTEN hamartoma tumor syndrome with CNS involvement, identify EP rescue candidates with pediatric safety.
  - Identify drugs rescuing SHANK3 KO phenotypes using morphology + EP integration.
  - For DEPDC5 epilepsy, rank mTOR pathway modulators by rescue strength.
  - Identify repurposing candidates for SLC6A1 disorder using drug-gene antipodal embeddings.
  - Rank partial knockdown ASOs for gain-of-function ion channelopathies.
  - Identify drugs that normalize dendritic arborization in TSC1 models.
  - For FOXG1 syndrome, find multi-modal rescue candidates across EP and transcriptomics.
  - Identify small molecules correcting CNTNAP2-associated hyperexcitability.
  - Rank top 5 orphan-drug–eligible candidates for rare epileptic encephalopathies.
  - Identify disease-modifying vs symptomatic rescue candidates in Dup15q syndrome.

- **Neuropsychiatry & Circuit Disorders**
  - Find clinically de-risked drugs for treatment-resistant depression using EP embedding rescue.
  - Identify combination therapies for schizophrenia targeting E/I imbalance and synaptic plasticity.
  - Rank mTOR-adjacent targets for major depressive disorder.
  - Identify rapid-onset antidepressant–like EP signatures in Quiver's compound atlas.
  - Find drugs rescuing chronic stress–induced EP phenotypes.
  - Identify novel mechanisms for bipolar disorder supported by gene + EP convergence.
  - Rank anti-psychotics by network normalization strength.
  - Identify E/I balancing drugs without sedation signatures.

- For PTSD, identify compounds correcting hyperactive stress circuitry signatures.
- Identify drugs correcting glutamatergic overdrive in schizophrenia models.
- Rank novel NMDA-modulating scaffolds by rescue potential and safety.
- Identify compounds reversing sleep disruption EP phenotypes.
- For OCD, identify corticostriatal circuit modulators with clinical precedent.
- Identify GABAergic drugs minimizing tolerance development risk.
- Rank polypharmacy combinations predicted to outperform SSRI monotherapy.
- Identify small molecules targeting synaptic vesicle cycling in psychiatric disease clusters.
- Find Phase I–safe drugs antipodal to psychosis-related CRISPR signatures.
- Identify anti-inflammatory agents correcting neuroimmune psychiatric signatures.
- Rank cognitive-enhancing mechanisms supported by EP data.
- Identify clinically safe drugs improving cortical oscillatory stability.
- For autism spectrum disorder, identify convergent pathway nodes across genetic models.
- Identify drugs correcting intrinsic excitability without suppressing plasticity.
- Rank lithium-adjacent mechanisms by EP rescue performance.
- Identify BBB-penetrant kinase inhibitors modulating psychiatric EP clusters.
- Find drugs rescuing long-term adaptation deficits in depressive neuron models.

- **Neurodegeneration**
  - Find clinically de-risked drugs for early Alzheimer’s disease using multi-modal rescue ranking.
  - Identify small molecules correcting tauopathy EP signatures.
  - Rank drugs reversing synaptic collapse in Parkinson’s disease models.
  - Identify mTOR-independent autophagy enhancers for neurodegeneration.
  - For ALS, rank compounds stabilizing motor neuron firing.
  - Identify combination strategies for Huntington’s disease.
  - Rank mitochondrial stabilizers by EP rescue strength.
  - Identify drugs correcting network desynchronization in dementia.
  - For FTD (MAPT mutation), identify antipodal drug signatures.
  - Identify compounds restoring adaptation metrics in aged neuron models.
  - Rank small molecules by predicted disease-modifying potential vs symptomatic.
  - Identify drugs correcting calcium dysregulation in neurodegeneration clusters.
  - Rank inflammation-modulating drugs by rescue in mixed neuron-glia models.
  - Identify drugs reducing pathological bursting in dopaminergic neurons.
  - For APOE4-associated dysfunction, identify network-normalizing compounds.
  - Rank top 5 FDA-approved drugs for repurposing in early AD.
  - Identify BBB-penetrant autophagy enhancers with human safety data.
  - For synucleinopathy, identify drugs reversing synaptic vesicle deficits.
  - Rank disease-modifying vs symptomatic candidates in Parkinson’s.
  - Identify polypharmacology strategies correcting multi-pathway dysfunction.

- **Pain & Ion Channel Disorders**
  - Identify small molecules targeting Nav1.7/Nav1.8 with minimal cardiovascular risk.
  - Rank ASO vs small molecule approaches for chronic pain.
  - Identify repurposing candidates for small fiber neuropathy.
  - Rank Phase II–safe sodium channel modulators by EP rescue.
  - Identify BBB-penetrant pain drugs with minimal sedation signature.

- Find drugs antipodal to CRISPR Nav1.7 GOF phenotype.
- Identify non-opioid analgesics supported by EP + genetics convergence.
- Rank kinase inhibitors modulating nociceptive neuron firing.
- Identify drugs correcting dorsal root ganglion hyperexcitability.
- Rank compounds with analgesic potential and minimal tolerance liability.
- Identify small molecules mimicking protective SCN9A LOF phenotype.
- Rank pain drugs by safety margin in human neuron EP data.
- Identify dual-mechanism pain combinations minimizing CNS depression.
- Identify ion channel modulators predicted to reduce chronic neuropathic pain.
- Rank clinically de-risked compounds for diabetic neuropathy.

- **Platform-Level / Multi-Disease Strategic Prompts**
  - Identify cross-disease convergent targets across epilepsy, ASD, and schizophrenia.
  - Rank top 10 CNS programs by Phase 2 probability uplift using Quiver validation.
  - Identify combination strategies maximizing rescue across 3 neurodevelopmental disorders.
  - Rank CRISPR suppressor hits by druggability and BBB feasibility.
  - Identify small molecules with multi-disease applicability based on embedding similarity.
  - Rank top 5 ASO programs ready for IND based on rescue + safety scoring.
  - Identify drug repurposing candidates with pediatric tolerability.
  - Rank novel scaffolds for CNS targeting using Quiver + chemoinformatics.
  - Identify drugs correcting both EP and morphology phenotypes.
  - Rank top 10 white-space CNS targets with low competitive density.
  - Identify combination therapies maximizing rescue with minimal toxicity signatures.
  - Rank candidates by composite rescue + clinical de-risking score.
  - Identify next experiments required to confirm top 3 ranked drugs.
  - For a $25M budget, prioritize 3 CNS programs based on rescue + feasibility.
  - Deliver a final prioritized list of clinically de-risked CNS programs across 5 diseases with mechanistic rationale and experimental plan.

- **ASO Sequence–Level Generation Prompts**
  - Design 5 gapmer ASO sequences targeting TSC2 mRNA that achieve 50–70% knockdown in human cortical neurons, maximize rescue of the TSC2 KO EP phenotype, and minimize predicted innate immune activation.
  - Generate allele-specific ASO sequences selectively suppressing the SCN8A R1872Q gain-of-function mutation without affecting wild-type transcript.
  - Design partial knockdown ASOs for MECP2 duplication syndrome targeting transcript regions that produce ~40% reduction while preserving physiological expression.
  - Generate splice-modulating ASOs that reduce inclusion of pathogenic exon variants in SCN2A while maintaining functional channel expression.
  - For SYNGAP1 haploinsufficiency, design ASOs targeting repressor elements to increase expression via antisense-mediated translational upregulation.
  - Design CNS-optimized ASO sequences targeting mTOR that reduce pathway hyperactivation but avoid full pathway suppression toxicity.
  - Generate 10 ASO sequences targeting Nav1.7 (SCN9A) optimized for intrathecal delivery, predicted BBB penetration, and minimal cardiotoxicity.
  - Design antisense oligonucleotides targeting GRIN2B gain-of-function variants using SNP-discriminating chemistry.
  - Generate ASOs targeting PTEN for controlled partial suppression, avoiding oncogenic liability predicted from transcriptomic models.

- Design exon-skipping ASOs for a TSC1 truncating mutation to restore reading frame and evaluate predicted EP rescue.
- Generate ASO candidates targeting SHANK3 transcripts that correct hyperexcitability signatures in excitatory neurons.
- Design multi-target ASOs capable of co-modulating two convergent pathway genes implicated in ASD cluster embeddings.
- Generate ASO sequences targeting DEPDC5 that reduce mTORC1 hyperactivation while preserving autophagy.
- Design allele-specific ASOs targeting CACNA1A missense mutation without suppressing total calcium channel expression.
- Generate antisense candidates targeting FOXG1 overexpression while minimizing developmental neurotoxicity signatures.
- Design chemically modified ASOs (2'MOE, LNA mixmer, PMO variants) targeting TSC2 and rank by predicted CNS stability and toxicity.
- Generate ASO sequences targeting SCN1A regulatory elements to enhance expression in inhibitory neurons.
- Design intron-targeting ASOs that modulate alternative splicing in CDKL5 to restore functional protein production.
- Generate 5 ASOs targeting KCNQ2 gain-of-function variants predicted to normalize firing rate without inducing hypoexcitability.
- Design ASOs targeting inflammatory mediator IL-1β transcripts implicated in neuroinflammation EP signatures.
- Generate allele-selective ASOs targeting mutant HTT transcript (Huntington's disease) while preserving wild-type allele.
- Design ASOs targeting MAPT (tau) that reduce pathogenic isoform expression but preserve total tau needed for microtubule stability.
- Generate antisense sequences targeting UBE3A antisense transcript (UBE3A-ATS) to reactivate paternal allele in Angelman syndrome.
- Design combinatorial ASO strategies targeting both SCN2A and downstream excitability modulators predicted to outperform monotherapy.
- For a given disease cluster (epilepsy, ASD, or TSC), generate a ranked list of 10 ASO sequences including:
  - Exact nucleotide sequences (5'→3')
  - Predicted knockdown %
  - Off-target risk score
  - Immunogenicity risk
  - CNS stability prediction
  - EP rescue probability score
  - Recommended next validation experiments